Correction to Hormesis as a Geometric Necessity of Bounded Adaptive Systems

Proposed author corrigendum for DOI 10.1177/15593258261469171

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This proposed correction concerns Murray DJ, *Dose Response*. 2026;24(3):15593258261469171, doi:10.1177/15593258261469171. It corrects the mathematical necessity, peak-location, finite-change attenuation, approximation and zone-width statements. It does not constitute an accepted or issued journal correction.

Necessity and baseline normalization. In Sections 3.3, 7.2, 7.5 and the Conclusion, activation before toxicity and greater high-dose damage are insufficient to guarantee a positive-dose beneficial interval. The adopted Möbius composition law supplies an additive rapidity coordinate; it does not choose the relative amplitude or slope of the two dose-dependent increments. Section 2.2 correctly states that Aczél's general theorem alone does not select that coordinate, and this clarification is retained.

For the published primary model with equal linear-dose logistic slope s, repair amplitude α, damage amplitude $\beta > \alpha$, and $0 < D_a < D_t$, set $a = e^{sD_a}$ and $t = e^{sD_t}$. A beneficial interval relative to the actual zero-dose control exists if and only if

$$\frac{\alpha a}{(1+a)^2} > \frac{\beta t}{(1+t)^2}.$$

Zero-dose logistic tails should be subtracted from the rapidity increments, or the response should be divided by its actual zero-dose value, so the control is exactly one. The dose-dependent peak is unchanged by this normalization. With $s = 1, D_a = 4, D_t = 5, \alpha = 0.3, \beta = 1$, the stated threshold ordering and high-dose inequality hold, but the response decreases throughout the positive-dose domain. The geometric-necessity claim and the corresponding mathematical exclusion of a monotonically harmful endpoint must therefore be withdrawn at their stated generality.

Peak location. Section 5.1, Prediction 1, Section 7.6, the Conclusion and Appendix A.2 should not state a universal peak near $2D_a$. For the equal-slope primary model, an interior peak exists under the condition above and is exactly

$$D_* = \frac{1}{s}\log\left[\frac{\sqrt{\alpha a}\,t - \sqrt{\beta t}\,a}{\sqrt{\beta t} - \sqrt{\alpha a}}\right].$$

The peak depends on both amplitudes, both thresholds and the slope. Unequal slopes require their own stationary equation. The later multiplicative Hill response has a different exact law and is not an algebraic replacement for the published linear-dose rapidity model.

Attenuation and finite changes. Sections 4.2–4.3, Predictions 3 and 5, related amplitude conclusions and Appendix A.3 extend a local metabolic-control approximation to finite changes. The summation theorem constrains the sum of local logarithmic coefficients. It neither forces equal control

coefficients nor determines a finite doubling. In the explicitly chosen equal-resistance series model, increasing one rate by a fraction A_{mol} gives

$$A_{\mathrm{func}} = \frac{A_{\mathrm{mol}}}{n + (n-1)A_{\mathrm{mol}}}, \qquad \eta = n + (n-1)A_{\mathrm{mol}}.$$

The approximation $\eta \simeq n$ is its small-change limit. Doubling one rate gives a flux increase of $1/3$ for two equal steps and $1/5$ for three, rather than $1/2$ and $1/3$. Other kinetic models give other finite responses, so step counts alone do not fix the observed attenuation.

The published comparisons with broad hormesis features and activation–toxicity ordering do not validate the withdrawn universal peak or finite-amplitude statements. The bounded-composition framework and conditional response models remain available. The companion mathematical amendment provides complete proofs, exact model-specific onset, peak and return-to-control laws, and independent reproducibility checks. The journal's handling and final wording of a formal corrigendum remain subject to editorial review.

Coincident thresholds and locally linear harm. Section 7.2's general near-coincident-threshold guarantee is withdrawn. Identical repair and damage sigmoids with equal midpoint and greater damage give no beneficial interval. In the common-slope logistic model at fixed amplitudes, sufficiently small threshold gaps also give none. Different sigmoid shapes require their own amplitude–slope conditions. A negative linear low-dose term is permitted in the common-slope logistic model and in the additive Hill model at exponent one. The composition law therefore does not exclude locally linear harm; this does not identify a globally proportional dose law.

Approximation bounds and zone width. Section 3.4 and Appendix A.1 require a full calibration and error definition for the product shortcut. Repair-only saturation matching and tanh concavity do not establish a conservative bound for the combined repair–damage response. The full product can overestimate the exact response even with a positive actual baseline. The uniform conservative interpretation and the unqualified error bound are withdrawn. The peak and zone-width ranges in Section 5.2, Table 2 and Appendix A.2 also depend on an incompletely stated steepness and control protocol. One compatible reconstruction, with each logistic slope set to $4/D_j$, reproduces Appendix A.2's range 1.69–2.43; this does not identify the original unstated protocol uniquely, and other slopes give other peaks. A detectable zone must use the actual control, a declared positive threshold and both threshold crossings. No universal 8–80-fold width follows from bounded composition.

Finite amplitudes and control. This also amends the peak-range, mean and aggregate-agreement claims in the abstract, Section 5.3 equation (8), Section 6.4 and Table 2. For the stated molecular changes $A_{\mathrm{mol}} \in [0.8, 1.5]$ and $n \in \{2, 3\}$, the corrected series-model flux factors span 117.4–142.9% of control. These are conditional finite flux ceilings, not a predicted distribution of hormesis peaks. A normalized bounded viability with actual control p_0 is below $1/p_0$ at finite rapidity, so its amplitude also requires an independently measured baseline and a flux-to-endpoint bridge.

References

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